Lab 02 — The Desk's Steps

One thing today: build the desk's seven steps from a ticker to the filings, the facts and the ratios, with no model yet, and match every number to the 10-K. (45 minutes, after the teaching and the demo.)

You arrive with: your setup checked in Lab 01, the paper's Section 3.1 and Figure 2 read, the report audit read. No model key is needed today; Thursday needs one.

Open — the paired review (10 minutes)

1. **Tools alive.** Terminal open in your work folder; `check_env.py` from Lab 01 prints your User-Agent; your AI agents answer in that folder. Nothing typed into them yet.
2. **Defend.** Tell your partner whether Figure 1 is a workflow or an agent, in one sentence, with the reason. Partner asks one question.
3. **Gap check.** On paper: what is a 10-K, what is a 10-Q, what is the 8-K's Exhibit 99.1, and what is XBRL? Write the one you could not explain.

D — the question, the same for everyone

> Given only a ticker, can your code find the company's filings, read its statements as data, and compute the core ratios, every number matching the 10-K?

R — what you work from

Input	Where
The seven steps	the instructor's demo on the screen, one step at a time
The filings and the data	SEC EDGAR: the ticker file, the submissions API, the documents, the XBRL company facts (the data page and the Week 2 page link Costco's)

Input	Where
SEC access	EDGAR asks every program for a User-Agent with your name and email, in your `.env`
The known answers	the instructor's run on the screen, read from what Costco filed last; each number judged by its filing (V below)

I — build it from the instruction

1. In your AI coding agent, in your work folder, send this request word for word:

> I am building the data side of an equity-research desk in Python, following Section 3.1 of Francis & Zhang (2026); the model roles come later. Read my SEC User-Agent as SEC_USER_AGENT from the .env file here and send it with every request. The program takes a ticker, and optionally --peers with a list of tickers, --price with a number, and --step with a number to run one step only. Nothing in the code names a company. It runs seven steps and prints each one as it goes. First, resolve the ticker to its CIK from SEC's company_tickers.json. Second, list the company's filings from the EDGAR submissions API and pick the latest 10-K, the latest 10-Q and the latest 8-K whose document reports Item 2.02, the earnings release; print each one's accession number and the "Accepted" time shown on its EDGAR index page. Third, download the three documents, taking the 8-K's Exhibit 99.1 for the release, convert each to plain text and save them in a folder named by company and the date and time of the run. Fourth, fetch the XBRL company facts and, for the latest fiscal year, print total revenue, operating income, net income, diluted EPS, stockholders' equity, long-term debt and its current portion, cash from operations and additions to property and equipment, each with its tag, period end and filing; when a concept has several candidate tags, choose the one whose latest year is newest, and read every concept for the same fiscal year as revenue, with the prior year beside it. Fifth, a calculator in code that prints, with its arithmetic, revenue growth, operating margin, return on average equity, debt to equity with the current portion included, free cash flow as cash from operations minus additions to property and equipment, and P/E from --price. Sixth, from the release text, print the lines that state the quarter's total revenue and diluted EPS. Seventh, for each peer, the same facts and the operating margin on the same definition. Save the facts beside the texts. When you are done, tell me the exact command to install any packages and the exact command to run it on a ticker with peers and a price.

Rewrite it in ASD-STE100 yourself before you send it, as Week 1 taught; a sample is here.

2. Run the install command your agent gives you, then the run command: `COST --peers WMT,TGT,BJ --price P`, where P is the close the instructor's run used. *Expect:* seven steps print, in the order the demo showed, ending with a saved folder.
3. **While it runs:** open a second AI chat and work one item from your gap list. Never collected.
4. Something not working? Debug with your AI: paste the exact command and the exact error text.

V — prove it on the known answer

The numbers are the instructor's run; the filing judges each one.

Step	The check
1	the CIK on Costco's EDGAR filing list
2	each filing's index page shows the same accession number and "Accepted" time; the earnings 8-K reports Item 2.02, and a newer 8-K that does not is skipped
3	each saved text reads as its filing, the release as Exhibit 99.1
4	each figure stands on its line of the 10-K's statements; every period end equals the 10-K's fiscal year end; each tag is the one that holds the latest year
5	each ratio redone by hand from step 4's printed inputs, on the report audit's definitions
6	the two lines appear word for word in the release
7	each peer's revenue and operating income stand in its own latest 10-K; every margin on one definition

Mark each step right or wrong, by hand, against the filings and the instructor's run. Your partner checks two marks. A revenue whose period end falls a year before the 10-K's fiscal year end came from an old tag; a margin whose two inputs show two period ends divided one year by another. Both are the tag trap, and both are yours to fix.

The filings as data — Learn on your own

1. What is XBRL (every number in a filing tagged with a concept, a period and a unit, served by EDGAR as one file per company)?
2. How does it work here (which tag your reader chose for revenue, and why the period matters)?
3. Explain it: to your partner, why a filer's tag can change between years and what your code does about it.

Reflect

Explain to your partner: 1. which step you would trust least on a company you have never seen, and why; 2. what the four roles will need from these steps on Thursday.

Before Session 3 — do these: the list on the Week 2 page, about 45 minutes.

Next: Lab 03 adds the four roles with your model key: the app, and its first report.